

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

- Technical Requirement ID: TR-CRM-001
- Related Functional Requirement(s): FR-CRM-001, FR-CRM-002, FR-CRM-003
- Objective: Design and implement a customer profile management system with CRUD operations and email validation.
- Target Cluster Configuration:
 - Cluster Name: Customer Relationship Management (CRM) Cluster
 - Databricks Runtime Version: 9.1 LTS
 - Node Type: Standard_DS3_v2
 - Driver Node: Standard_DS3_v2
 - Worker Nodes: 2-5 nodes (autoscaling)
 - Autoscaling: Enabled
 - Auto Termination: Enabled (30 minutes)
 - Libraries Installed:
 - Email validation library (e.g., Apache Commons Validator)
 - Databricks Spark library (for data processing)
- Source Data Details:
 - Dataset Name: Customer Profiles
 - Location: `/customer/profiles` (Parquet format)
 - Format: Parquet
 - Description: Stores customer profile information
- Target Data Details:
 - Output Name: Customer Profile Output
 - Location: `/customer/output` (JSON format)
 - Format: JSON
 - Description: Output file containing customer profile information
- Job Flow / Pipeline Stages:
 - Stage 1: Create/Update Customer Profile
 - Stage 2: Validate Email Address and Phone Number

- Stage 3: Send Welcome Email (upon new customer creation)
- Data Transformations / Business Logic:
 - Step 1: Validate Email Address
 - Description: Validate the format of the entered email address
 - Transformation Logic: Use Apache Commons Validator library to check email format
 - Step 2: Validate Phone Number
 - Description: Validate the format of the entered phone number
 - Transformation Logic: Use regular expressions to check phone number format
 - Step 3: Create/Update Customer Profile
 - Description: Create or update customer profile information in the database
 - Transformation Logic: Use Databricks Spark to process and store customer profile data
- Error Handling and Logging:
 - Log all errors and exceptions to a designated error log file
 - Implement retry mechanisms for email sending and data processing tasks
 - Display user-friendly error messages to users upon encountering errors
- Technical Requirement ID: TR-CRM-002
- Related Functional Requirement(s): FR-CRM-004
- Objective: Implement email and phone number validation for customer contact information.
- Data Transformations / Business Logic:
 - Step 1: Validate Email Address Format
 - Description: Check if the email address is in the correct format
 - Transformation Logic: Use Apache Commons Validator library to check email format
 - Step 2: Validate Phone Number Format
 - Description: Check if the phone number is in the correct format
 - Transformation Logic: Use regular expressions to check phone number format
- Technical Requirement ID: TR-CRM-003
- Related Functional Requirement(s): FR-CRM-005
- Objective: Implement a welcome email sending system for new customers.
- Job Flow / Pipeline Stages:
 - Stage 1: Generate Welcome Email
 - Stage 2: Send Welcome Email to Customer
- Data Transformations / Business Logic:

- Step 1: Generate Welcome Email
 - Description: Generate a welcome email using a predefined template
 - Transformation Logic: Use a templating engine (e.g., Jinja2) to populate the email template with customer information
- Step 2: Send Welcome Email
 - Description: Send the welcome email to the customer's email address
 - Transformation Logic: Use a mail server (e.g., SMTP) to send the email
- Error Handling and Logging:
 - Log all email sending errors to a designated error log file
 - Implement retry mechanisms for email sending tasks